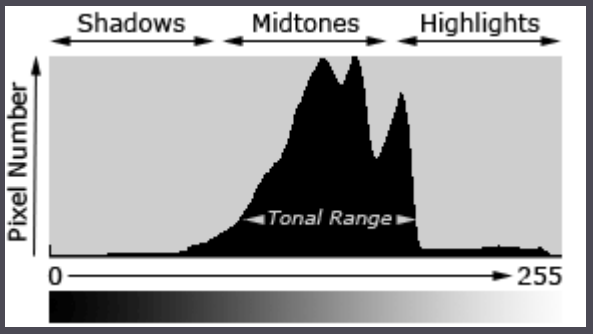
CAMERA HISTOGRAMS: TONES & CONTRAST

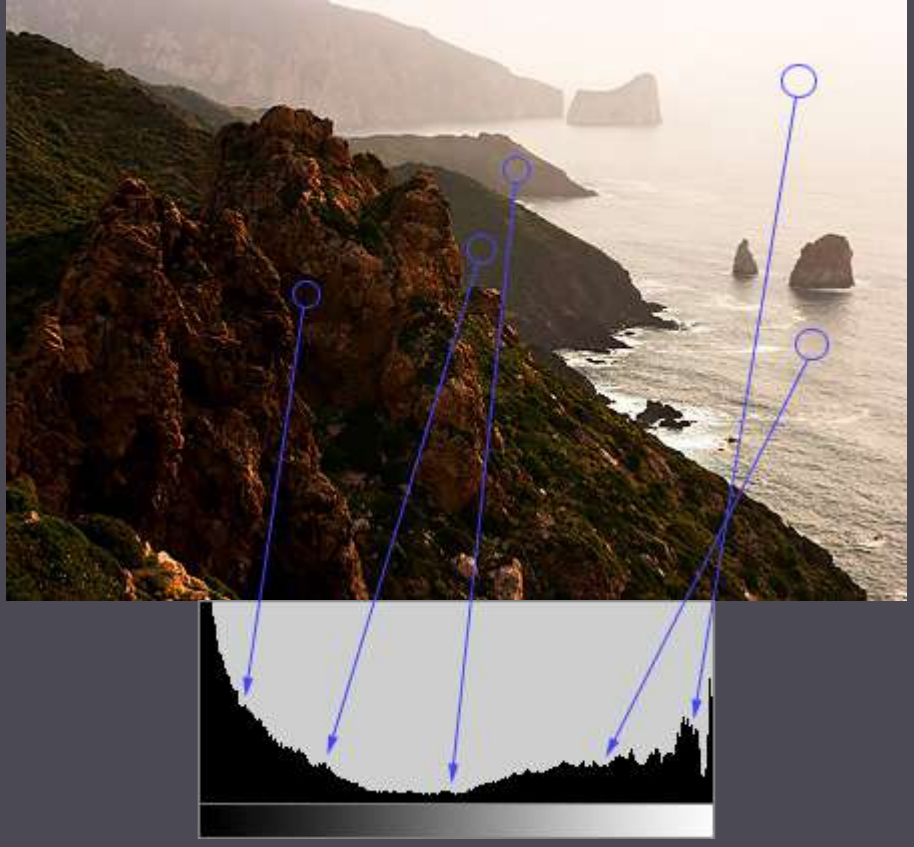
Understanding image histograms is probably the single most important concept to become familiar with when working with pictures from a digital camera. A histogram can tell you whether or not your image has been properly exposed, whether the lighting is harsh or flat, and what adjustments will work best. It will not only improve your skills on the computer, but as a photographer as well.

Each pixel in an image has a color which has been produced by some combination of the primary colors red, green, and blue (RGB). Each of these colors can have a brightness value ranging from 0 to 255 for a digital image with a bit depth of 8-bits. A RGB histogram results when the computer scans through each of these RGB brightness values and counts how many are at each level from 0 through 255. Other types of histograms exist, although all will have the same basic layout as the histogram example shown below.

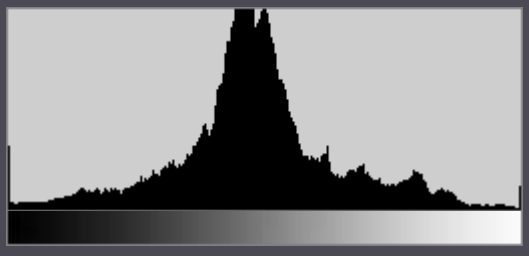


TONES

The region where most of the brightness values are present is called the "tonal range." Tonal range can vary drastically from image to image, so developing an intuition for how numbers map to actual brightness values is often critical—both before and after the photo has been taken. There is no one "ideal histogram" which all images should try to mimic; histograms should merely be representative of the tonal range in the scene and what the photographer wishes to convey.



The above image is an example which contains a very broad tonal range, with markers to illustrate where regions in the scene map to brightness levels on the histogram. This coastal scene contains very few midtones, but does have plentiful shadow and highlight regions in the lower left and upper right of the image, respectively. This translates into a histogram which has a high pixel count on both the far left and right-hand sides.

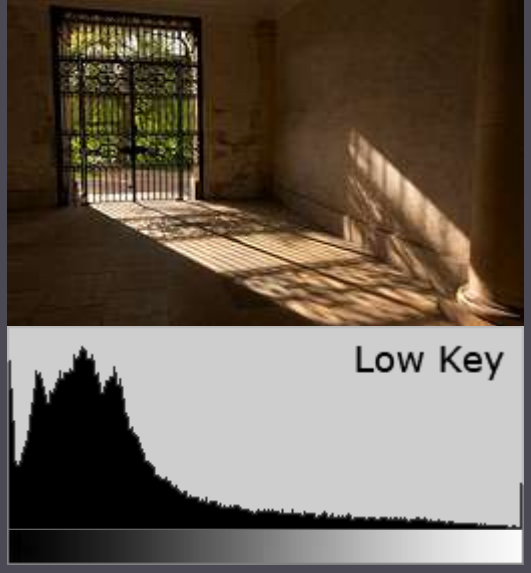


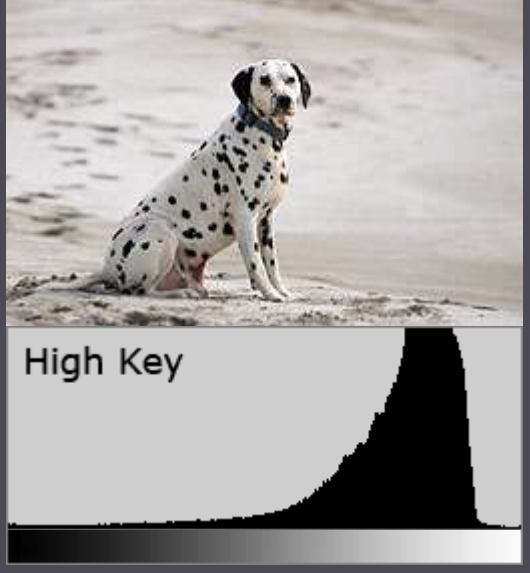


Lighting is often not as extreme as the last example. Conditions of ordinary and even lighting, when combined with a properly exposed subject, will usually produce a histogram which peaks in the center, gradually tapering off into the shadows and highlights. With the exception of the direct sunlight reflecting off the top of the building and off some windows, the boat scene to the right is quite evenly lit. Most cameras will have no trouble automatically reproducing an image which has a histogram similar to the one shown in the boat example.

HIGH AND LOW KEY IMAGES

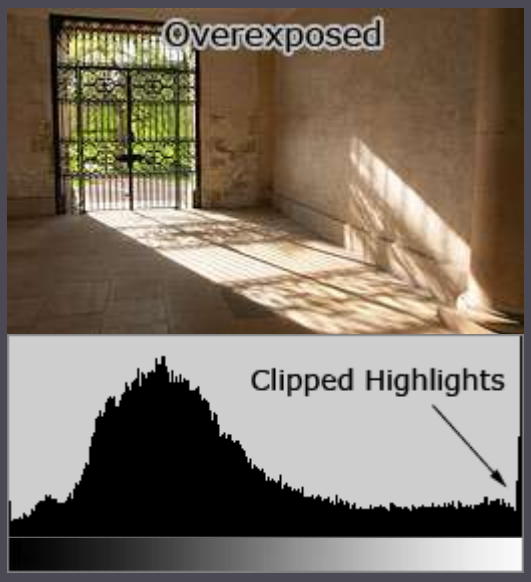
Although most cameras will produce midtone-centric histograms when in an automatic exposure mode, the distribution of peaks within a histogram also depends on the tonal range of the subject matter. Images where most of the tones occur in the shadows are called "low key," whereas with "high key" images most of the tones are in the highlights.

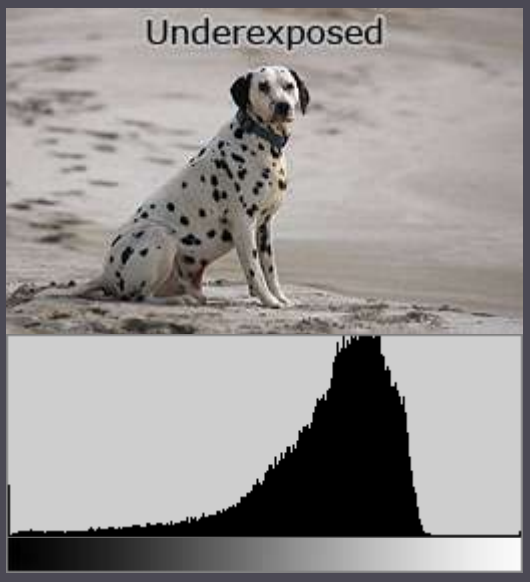




Before the photo has been taken, it is useful to assess whether or not your subject matter qualifies as high or low key. Since cameras measure reflected as opposed to incident light, they are unable to assess the absolute brightness of their subject. As a result, many cameras contain sophisticated algorithms which try to circumvent this limitation, and estimate how bright an image should be. These estimates frequently result in an image whose average brightness is placed in the midtones. This is usually acceptable, however high and low key scenes frequently require the photographer to manually adjust the exposure, relative to what the camera would do automatically. A good rule of thumb is that you will need to manually adjust the exposure whenever you want the average brightness in your image to appear brighter or darker than the midtones.

The following set of images would have resulted if I had used my camera's auto exposure setting. Note how the average pixel count is brought closer to the midtones.





Most digital cameras are better at reproducing low key scenes since they prevent any region from becoming so bright that it turns into solid white, regardless of how dark the rest of the image might become as a result. High key scenes, on the other hand, often produce images which are significantly underexposed. Fortunately, underexposure is usually more forgiving than overexposure (although this compromises your signal to noise ratio). Detail can never be recovered when a region becomes so overexposed that it becomes solid white. When this occurs the highlights are said to be "clipped" or "blown."

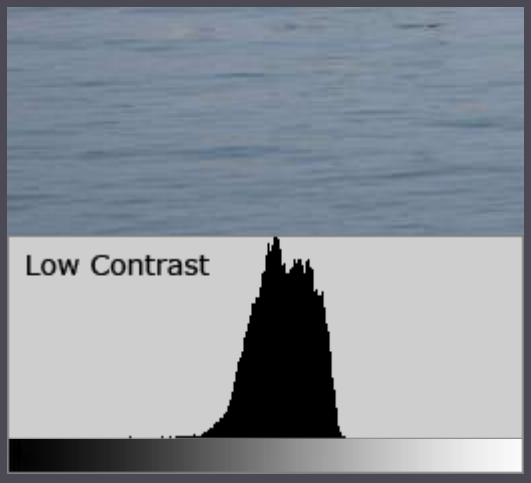


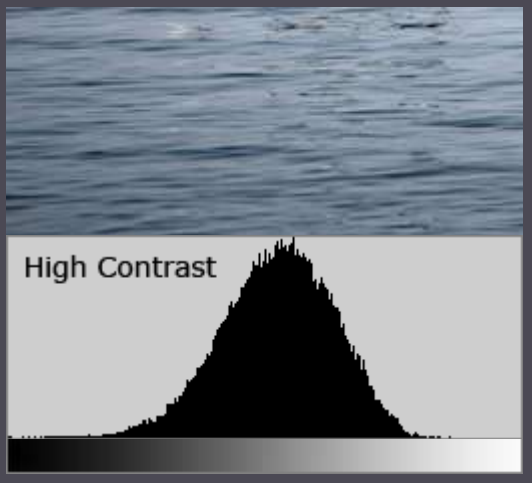


The histogram is a good tool for knowing whether clipping has occurred since you can readily see when the highlights are pushed to the edge of the chart. Some clipping is usually ok in regions such as specular reflections on water or metal, when the sun is included in the frame or when other bright sources of light are present. Ultimately, the amount of clipping present is up to the photographer and what they wish to convey.

CONTRAST

A histogram can also describe the amount of contrast. Contrast is a measure of the difference in brightness between light and dark areas in a scene. Broad histograms reflect a scene with significant contrast, whereas narrow histograms reflect less contrast and may appear flat or dull. This can be caused by any combination of subject matter and lighting conditions. Photos taken in the fog will have low contrast, while those taken under strong daylight will have higher contrast.

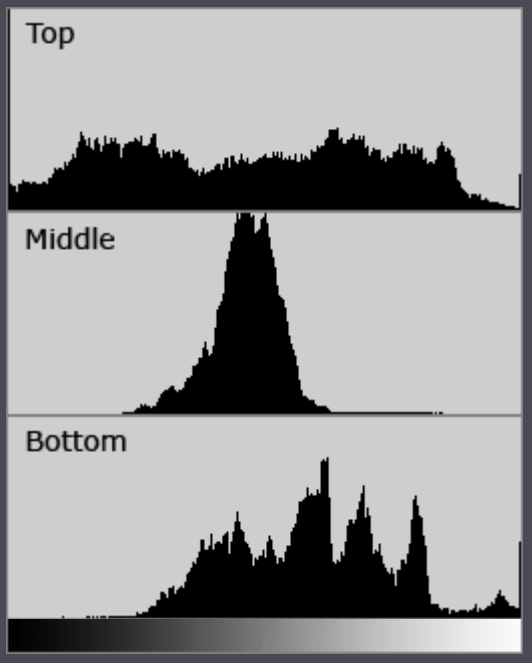




Contrast can have a significant visual impact on an image by emphasizing texture, as shown in the image above. The high contrast water has deeper shadows and more pronounced highlights, creating texture which "pops" out at the viewer.

Contrast can also vary for different regions within the same image due to both subject matter and lighting. We can partition the previous image of a boat into three separate regions—each with its own distinct histogram.





The upper region contains the most contrast of all three because the image is created from light which does not first reflect off the surface of water. This produces deeper shadows underneath the boat and its ledges, and stronger highlights in the upward-facing and directly exposed areas. The middle and bottom regions are produced entirely from diffuse, reflected light and thus have lower contrast; similar to if one were taking photographs in the fog. The bottom region has more contrast than the middle—despite the smooth and monotonic blue sky—because it contains a combination of shade and more intense sunlight. Conditions in the bottom region create

more pronounced highlights, but it still lacks the deep shadows of the top region. The sum of the histograms in all three regions creates the overall histogram shown before.

For additional information on histograms, visit part 2 of this tutorial:
"Understanding Camera Histograms: Luminosity & Color"

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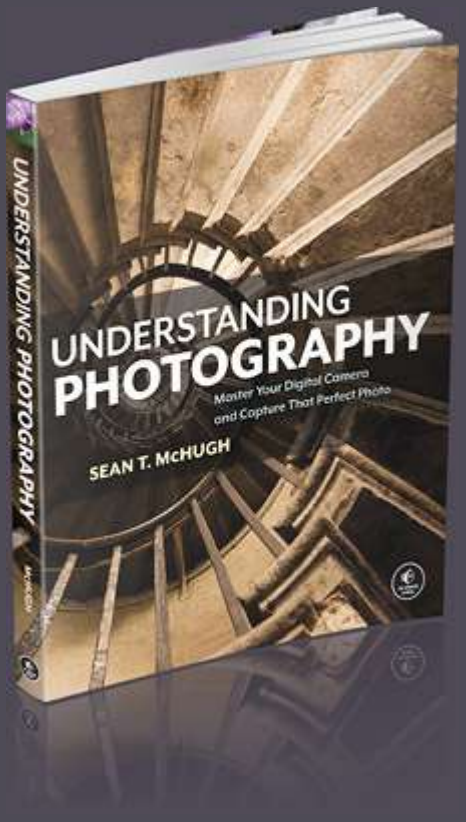
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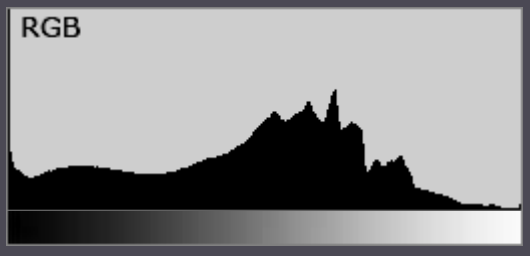
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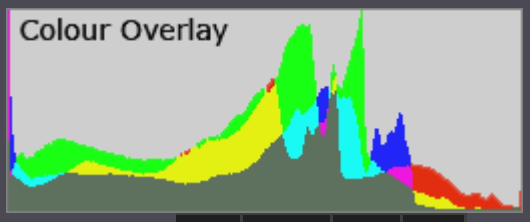


CAMERA HISTOGRAMS: LUMINOSITY & COLOR

This section is designed to help you develop a better understanding of how luminosity and color both vary within an image, and how this translates into the relevant histogram. Although RGB histograms are the most commonly used histogram, other types are more useful for specific purposes.

The image below is shown alongside several of the other histogram types which you are likely to encounter. Move your mouse over the labels at the bottom to toggle which type of color histogram is displayed. When you change to one of the color histograms a different image will be shown. This new image is a grayscale representation of how that color's intensity is distributed throughout the image. Pay particular attention to how each color changes the brightness distribution within the image, and how the colors within each region influence this brightness.





Choose: Red Green Blue ALL



Tap or mouseover image
to view as luminosity

LUMINANCE HISTOGRAMS

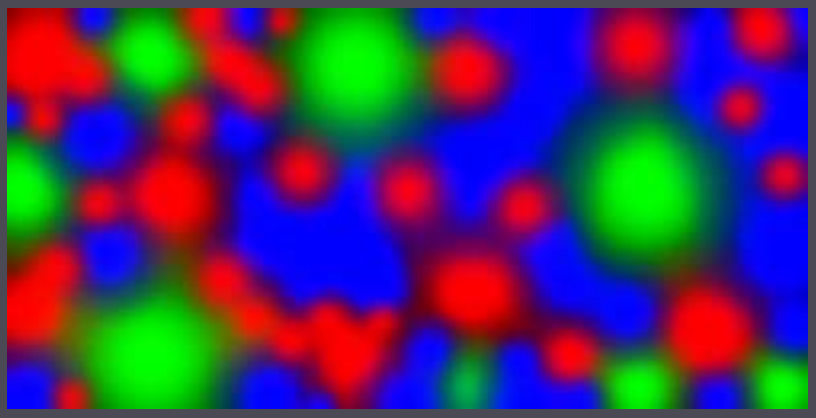
Luminance* histograms are more accurate than RGB histograms at describing the perceived brightness distribution or "luminosity" within an image. Luminosity takes into account the fact that the human eye is more sensitive to green light than red or blue light. View the above example again for each color and you will see that the green intensity levels within the image are most representative of the brightness distribution for the full color image. This also reflected by the fact that the luminance histogram also matches the green histogram more than any other color. Luminosity correctly predicts that the following stepped gradient gradually increases in lightness, whereas a simple addition of each RGB value would give the same intensity at each rectangle.

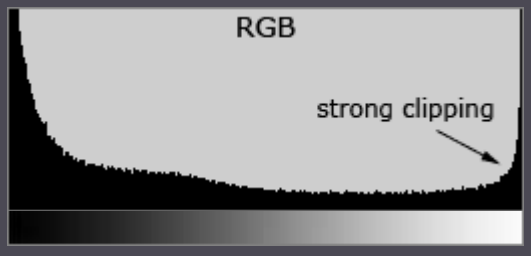


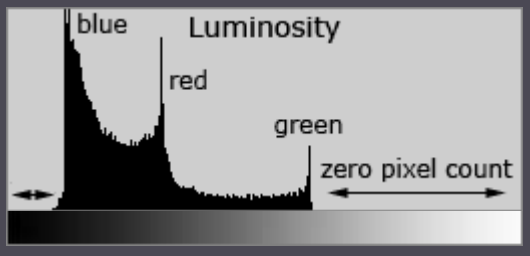
How is a luminance histogram produced? First, each pixel is converted so that it represents a luminosity based on a weighted average of the three colors at that pixel. This weighting assumes that green represents 59% of the perceived luminosity, while the red and blue channels account for just 30% and 11%, respectively. Move your mouse over "convert to luminosity" under the example image above to see what this calculation looks like when performed for each pixel. Once all pixels have been converted into luminosity, a luminance histogram is produced by counting how many pixels are at each brightness — identical to how a histogram is produced for a single color.

***Technical Note**: Strictly speaking, these should really be called "luminosity histograms." Unfortunately, the terms "luminance" and "luminosity" are often used interchangeably, including by Photoshop, even though each describes a different aspect of light intensity. Luminance refers to the absolute amount of light emitted by an object per unit area, whereas luminosity refers to the perceived brightness of that object by a human observer.

An important difference to take away from the above calculation is that while luminance histograms keep track of the location of each color pixel, RGB histograms discard this information. An RGB histogram produces three independent histograms and then adds them together, irrespective of whether or not each color came from the same pixel. To illustrate this point we will use an image which the two types of histograms interpret quite differently.







The above image contains many patches of pure color. At the interior of each color patch the intensity reaches a maximum of 255, so all patches have significant color clipping and only in that color. Even though this image contains no pure white pixels, the RGB histogram shows strong clipping—so much that if this were a photograph the image would appear significantly overexposed. This is because the RGB histogram does not take into account the fact that all three colors never clip in the same place.

The luminance histogram tells an entirely different story by showing no pixels anywhere near full brightness. It also shows three distinct peaks—one for each color that has become significantly clipped. Since this image contains primarily blue, then red, then least of all green, the relative heights clearly show which color belongs where. Also note that the relative horizontal position of each peak is in accordance with the percentages used in the weighted average for calculating luminance: 59%, 30%, and 11%.

So which one is better? If we cared about color clipping, then the RGB histogram clearly warns us while the luminance histogram provides no red flags. On the other hand, the luminance histogram accurately tells us that no pixel is anywhere near full black or white. Each has its own use and should be used as a collective tool. Since most digital cameras show only a RGB histogram, just be aware of its shortcomings. As a rule of thumb, the more intense and pure the colors are in your image, the more a luminance and RGB histogram will differ. Pay careful attention when your subject contains strong shades of blue since you will rarely be able to see blue channel clipping with luminance histograms.

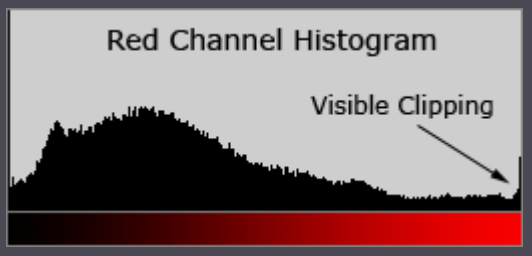
COLOR HISTOGRAMS

Whereas RGB and luminance histograms use all three color channels, a color histogram describes the brightness distribution for any of these colors individually. This can be more helpful when trying to assess whether or not individual colors have been clipped.

View Channel: Red | Green | Blue | ALL | Luminosity







View Histogram: RGB | Luminosity

The petals of the red flowers caught direct sunlight, so their red color became clipped, even though the rest of the image remained within the histogram. Regions where individual color channels are clipped lose all texture caused by that particular color. However, these clipped regions may still retain some luminance texture if the other two colors have not also been clipped. Individual color clipping is often not as objectionable as when all three colors clip, although this all depends upon what you wish to convey.

RGB histograms can show if an individual color channel clips, however they do not tell you if this is due to an individual color or all three. Color histograms amplify this effect and clearly show the type of clipping. Move your mouse over the labels above to compare the luminance and RGB histograms, to view the image in terms of only a single color channel, and to view the image luminance. Notice how the intensity distribution for each color channel varies drastically in regions of nearly pure color. The strength and purity of colors within this image cause the RGB and luminance histograms to differ significantly.

For additional information on histograms, visit part 1 of this tutorial:
"Understanding Camera Histograms - Tones and Contrast"

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